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LECTURE #1

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0:00 - 0:10

Introducing myself. Milan.

Purpose of this course: Basic scattering theory.

Theory of collisions in QM. Particle physics, chemical reactions,
light-matter interaction, scattering of light in waveguides.

This is not a Gaokao class. Slow pace of lecturing. Understanding, quality over quantity.
Assessment: Project + oral interview.

Check with your academic advisor if this course is recognized by your major.

Questions ?

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0:10-0:15

Classical particle in 1D (whole real axis). Newtonian + Hamiltonian formulation.

0:15-0:25

Quantization, Hamiltonian. Square integrable wavefunctions.
TDSE, TISE. Operators, expectation values.

0:25-0:35

Bound potentials:

Bound states. Quantization of energy levels.

Orthonormality and closure properties. Dirac delta function of position.

TDSE solved in eigenstate basis.

0:35-0:45

No potential:

Continuum states - not square integrable.

Orthonormality and closure properties.

Dirac delta functions of position and momentum.

TDSE solved in eigenstate basis.

0:45-0:55

Short ranged potentials:

Bound + continuum states.

Scattering boundary conditions for the continuum states.

Degeneracy of continuum states.

Orthonormality and closure properties.

Dirac delta functions of position and momentum.

TDSE solved in eigenstate basis.

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0:55 - 1:05

Break.

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1:05-1:15

Methods #1:

Basis set expansion method for bound states.

1:15-1:25

Methods #2:

Finite difference method for solving ODEs.

Application for continuum states (Neuhauser trick).

1:25-1:35

Scattering of wavepackets in the time domain.

TDSE, qualitative description of the scattering phenomenon.

Asymptotic condition, S-operator.

1:35-2:00

Description of the first part of the tutorial project:

Scattering of wavepackets in the time domain.

Resonance tunneling phenomenon discovered in hermitian wavepacket calculations.

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